

Linear Regression House Price Predictor

Task 1 Short Report

Objective

The objective of this task was to complete a basic machine learning workflow for house price prediction using Linear Regression.

The work included data loading, exploration, preprocessing, train-test split, model training, evaluation, visualization, and reporting.

Dataset and Setup

Dataset used: California Housing data uploaded as housing.csv.

Rows: 20,640. Columns after preprocessing: 13.

Target variable: MedHouseVal, representing median house value in units of 100,000 dollars.

The uploaded CSV was loaded, inspected, cleaned, and converted into numeric model-ready features.

Preprocessing Performed

- Missing numeric values were checked and handled using median filling where needed.
- Raw totals were converted into average room, bedroom, and occupancy features.
- The ocean_proximity categorical column was converted using one-hot encoding.
- The dataset was split into 16,512 training rows and 4,128 testing rows.

Exploratory Data Analysis

EDA Steps

I checked the dataset shape, column types, missing values, descriptive statistics, distributions, and correlations.

Distribution plots were used to understand skewed variables and the range of feature values.

A correlation heatmap was used to compare how each feature related to MedHouseVal.

Important EDA Findings

- Median income showed the strongest positive relationship with house value.
- Ocean proximity was important: inland locations were negatively related to house value.
- Latitude and longitude helped capture location-based differences in California prices.
- Some features were skewed, so transformations could improve future models.

Model Used

The model used for this task was LinearRegression from scikit-learn.

Linear Regression was chosen because it is simple, interpretable, and works well as a baseline model.

The feature set included income, location, house age, population, engineered average features, and encoded ocean-proximity columns.

Model Evaluation and Conclusion

Evaluation Metrics

MAE: 0.538797, which is about 53,880 dollars average absolute error.

RMSE: 0.812622, which is about 81,262 dollars with larger errors penalized more heavily.

R2 Score: 0.496070, meaning the model explains about 49.6 percent of the test-set variance.

Visual Evaluation

The predicted vs actual scatter plot showed that the model follows the overall trend but still has noticeable spread.

The residual plot showed remaining patterns, which means the relationship is not fully captured by a simple linear model.

Conclusion

The project successfully completed the required machine learning lifecycle from data loading to reporting.

Linear Regression provided a useful baseline, but the results show that house prices have nonlinear patterns.

Improvement Ideas

- Try Ridge Regression, Lasso Regression, Random Forest, or Gradient Boosting.
- Add cross-validation and stronger location-based feature engineering.
- Apply transformations to skewed variables and compare scaled vs unscaled performance.

The saved model file from the notebook can be used for simple predictions on new housing inputs.